Annotation Protocol and Standard Guidelines

1 FISWG Guidelines Overview and Application in Facial Description Origin and Purpose

The *Facial Identification Scientific Working Group* (FISWG) was established to develop scientifically sound, standardized guidelines for facial identification practices. These standards aim to ensure that morphological comparisons and facial descriptions are accurate, reproducible, and legally defensible in forensic and biometric contexts. FISWG guidelines provide structure for documenting and describing facial morphology, focusing on observable, measurable features rather than subjective or ethnically inferred traits. The standards are used globally across law enforcement, forensic laboratories, and biometric research institutions.

2 Key Principles Utilized in the Current Study

1. Objectivity and Neutrality
   Descriptions avoid speculative or interpretive language regarding ethnicity, emotional state, or personality. Instead, they emphasize observable anatomical details, maintaining forensic neutrality and compliance with ISO/IEC 19794-5:2011[2] for biometric facial image standards.

2. Hierarchical Structure of Facial Description
   Each subject's facial morphology is described following the top-to-bottom anatomical order recommended by FISWG:

   (a) General Demographics: Estimated age range, sex (if determinable), complexion, and overall facial form.

   (b) Stable Features: Bone-structured features that remain constant over time, e.g., forehead shape, brow ridge prominence, orbital depth, nose structure, cheekbone contour, jawline, and chin morphology.

   (c) Transient Features: Changeable elements such as hairstyle, beard or mustache, scars, moles, adornments (e.g., jewellery, glasses), and expressions.

   (d) Symmetry and Proportionality: Bilateral comparisons (left/right alignment of eyes, eyebrows, nostrils, and ears).

   (e) Overall Impression: Summary of distinguishing morphological identifiers for forensic comparison.

3. Terminology and Measurement Approach

   - Descriptive vocabulary adheres to FISWG Morphological Comparison Terminology (2021 Revision)[1], which standardizes terms like medium nasal bridge, oval face, almond-shaped eyes, defined malar contour, and broad chin base.
   - Quantitative measurement is not inferred; instead, relative proportional terminology ("moderate width," "slightly prominent," "well-defined") is used for reproducibility without instrument bias.

4. Reproducibility and Examiner Agreement
   To ensure inter-examiner consistency, all descriptions are formulated in a way that any trained analyst using FISWG guidelines would generate a *comparable morphological report.* Each description can therefore support *photo-to-photo verification, facial composite reconstruction, or anthropometric modeling.*

3 Forensic and Research Relevance

The FISWG-based framework ensures that facial documentation is:

- *Forensically admissible* (traceable to recognized scientific guidelines).
- *Empirically standardized*, reducing subjectivity across observers.
- *Applicable in AI/ML-based facial recognition research*, where consistent feature annotation is critical for dataset reliability.

This methodological alignment strengthens both the validity of human morphological analysis and the training accuracy of forensic facial recognition systems, forming a scientifically defensible bridge between traditional forensic anthropology and computational biometrics.

1. It refers to the forensic analysis that adheres to the requirements of admissibility, in accordance with the scientific guidelines. It has to be empirically standardized in such a way as to reduce variations in observations.
2. It is significant in AI/ML-based research studies on facial recognition, as uniform annotation of features is required to maintain the integrity of the datasets.
3. It connects the analysis of human morphology with the systems of forensic facial recognition, establishing a scientifically justifiable "bridge" between traditional anthropology and computer-based biometrics.

References

[1] Facial Identification Scientific Working Group (FISWG). Guidelines for facial comparison methods, 2021.

[2] International Organization for Standardization. Information technology – biometric data interchange formats – part 5: Face image data, 2011.